

```
# -*- coding: utf-8 -*-
```

```
"""
```

```
Created on Wed Sep 26 21:34:35 2018
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```
@author: jfowl
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```
"""
```

```
import numpy as np
```

```
import control as ctl
```

```
import matplotlib.pyplot as plt
```

```
this_clss = 'ME_426'
```

```
homework = 'HW_6'
```

```
# %% PROBLEM 2a FOR PROBLEM 3
```

```
def sys_model2a(tv):
```

```
    """Problem 2a"""
```

```
    x = []
```

```
    for t in tv:
```

```
        x.append(1 - np.exp(-5*t))
```

```
    return x
```

```
tvec2a = np.linspace(0, 2, 1000)
```

```
tf2a = ctl.tf([5], [1, 5, 0])
```

```
tout2a, cout2a = ctl.impulse_response(tf2a, tvec2a)
```

```
plt.figure(1, figsize=(22, 14))
```

```
plt.plot(tvec2a, sys_model2a(tvec2a), 'r', linewidth=4,  
         label='Analytic Solution')
```

```
plt.plot(tout2a, cout2a, '-.k', linewidth=2, label='Transfer Function Model')
```

```
plt.xlim([min(tvec2a), max(tvec2a)])
```

```
plt.ylim([min(cout2a), max(cout2a)])
```

```
plt.xlabel('Time (s)', fontsize=16)
```

```
plt.ylabel('C(t)', fontsize=16)
```

```
plt.title('C(t) vs. Time {}'.format(sys_model2a.__doc__), fontsize=18)
```

```
plt.legend()
```

```
plt.grid()
```

```
plt.savefig('{}_{}_{}_Plot.png'.format(this_clss, homework,  
                                         sys_model2a.__doc__))
```

```
plt.show()
```

```
# %% PROBLEM 2b FOR PROBLEM 3
```

```
def sys_model2b(tv):
```

```
    """Problem 2b"""
```

```
    x = []
```

```
    for t in tv:
```

```
        x.append(1 - np.exp(-20*t))
```

```
    return x
```

```
tvec2b = np.linspace(0, 2, 1000)
```

```
tf2b = ctl.tf([20], [1, 20, 0])
```

```

tout2b, cout2b = ctl.impulse_response(tf2b, tvec2b)

plt.figure(2, figsize=(22, 14))
plt.plot(tvec2b, sys_model2b(tvec2b), 'r', linewidth=4,
         label='Analytic Solution')
plt.plot(tout2b, cout2b, '-.k', linewidth=2, label='Transfer Function Model')
plt.xlim([min(tvec2b), max(tvec2b)])
plt.ylim([min(cout2b), max(cout2b)])
plt.xlabel('Time (s)', fontsize=16)
plt.ylabel('C(t)', fontsize=16)
plt.title('C(t) vs. Time {}'.format(sys_model2b.__doc__), fontsize=18)
plt.legend()
plt.grid()
plt.savefig('{}_{}_{}_Plot.png'.format(this_cls, homework,
                                       sys_model2b.__doc__))
plt.show()

```